Name: _____ **Roll number:** _____

Gradient boosting, numbers (regression) **Where you see a), b), c): circle the one right answer.**

The idea. We want to predict a number (this is called *regression*). We start with one guess for every point. Then we repeat: (1) see what is *left over* (our mistakes); (2) draw a tiny tree that predicts the leftover; (3) add a small part of it to the guess. Every round the guess gets better.

point i	1	2	3	4	5	6
x_i	1	2	3	4	5	6
y_i	1	2	3	8	10	12

A **stump** is the smallest tree: one question ("is $x \le t$?") and two leaves. Step size $\nu = \frac{1}{2}$: we add only half of each leaf value.

Formulas you need.
- first guess F_0 = average of y
- leftover $r = y - F$
- leaf value = average of the leftovers in that leaf
- new guess $F_{\text{new}} = F_{\text{old}} + \nu \times$ leaf value
- total leftover = $\sum r^2$ (square each leftover, then add them)

Step 1 First guess and leftover.

(a) The first guess F_0 is the average of y:

a) 5.5 **b)** 6 **c)** 7

(b) The leftover $r = y - F_0$ for points 1 to 6 is:

a) $-5, -4, -3, 2, 4, 6$ **b)** $5, 4, 3, -2, -4, -6$ **c)** $1, 2, 3, 8, 10, 12$

(c) A positive r means our guess is **a)** too high **b)** too low.

Step 2 Choose the split.

A good split puts points with *similar* leftovers in one leaf. Points that satisfy the split (for example $x \le 2.5$) go to the left leaf, the others to the right. Use your leftovers from Step 1(b) and write the ones that fall in each leaf.

split	left leaf	right leaf
$x \le 2.5$	$-5, -4$	$-3, 2, 4, 6$
$x \le 3.5$	$-5, -4, -3$	$2, 4, 6$
$x \le 4.5$	$-5, -4, -3, 2$	$4, 6$

Best split:

a) $x \le 2.5$ **b)** $x \le 3.5$ **c)** $x \le 4.5$

Why is it best? *All leftovers on its left are negative and all on its right are positive; the other splits mix signs.*

Step 3 Leaf values.

Leaf value = average leftover in the leaf. For the best split, (left, right) = **a)** $-4, +4$ **b)** $-12, +12$ **c)** $+4, -4$ **d)** $-2, +2$

Step 4 Add a small part.

The new guess $F_1 = F_0 + \nu \times$ leaf value, for points 1 to 6, is:

a) 4, 4, 4, 8, 8, 8 **b)** 2, 2, 2, 10, 10, 10 **c)** $-2, -2, -2, 2, 2, 2$

The new leftover $r = y - F_1$ is:

a) $-3, -2, -1, 0, 2, 4$ **b)** $-5, -4, -3, 2, 4, 6$ **c)** $3, 2, 1, 0, -2, -4$

Total leftover $\sum r^2$ (square each leftover, then add) goes from 106 to **a)** 34 **b)** 74 **c)** 106. **Why not add the whole leaf value ($\nu = 1$)?** *One tiny tree from six points is a rough guess; small steps let later trees fix it instead of locking its errors in.*

Step 5 Round 2.

The new leftovers are $-3, -2, -1, 0, 2, 4$. Choose the split the same way as in Step 2. The better split is **a)** $x \le 3.5$ **b)** $x \le 4.5$. **Why?** *The signs change between points 4 and 5, so $x \le 4.5$ gives leaf values -1.5 (left) and +3 (right).*

Think. (a) The second tree predicts the **a)** leftover **b)** original y. **Why?** *The first tree already explained the big pattern; the leftover is exactly what it missed.*

(b) AdaBoost passes its mistakes to the next tree as: **a)** weights on the points **b)** leftovers

Gradient boosting passes its mistakes to the next tree as: **a)** weights on the points **b)** leftovers

Name: _____ **Roll number:** _____

Gradient boosting, yes/no (classification) **Where you see a), b), c): circle the one right answer.**

The idea. The answer is now yes (1) or no (0). The model keeps a **score** F for every point (any number; $F = 0$ means fifty-fifty) and turns it into a **probability of yes** $p = \sigma(F)$. The loop is the same as on page 1: (1) find the leftover $r = y - p$; (2) draw a stump that predicts the leftover; (3) add a small part of it to the *score*.

point i	1	2	3	4	5	6
x_i	1	2	3	4	5	6
y_i	0	0	1	1	0	1

Same stumps as page 1. Step size $\nu = \frac{1}{2}$.

Formulas you need.
- probability of yes $p = \sigma(F) = \dfrac{1}{1+e^{-F}}$; so $F = 0$ gives $p = 0.5$
- first score $F_0 = \ln \dfrac{\text{number of 1s}}{\text{number of 0s}}$
- leftover $r = y - p$
- leaf value $= \sum r \,/\, \sum p(1-p)$, both sums over the points in that leaf
- new score $F_{\text{new}} = F_{\text{old}} + \nu \times$ leaf value
- $\sigma(-1) = 0.269$, $\sigma(0.5) = 0.622$

Step 1 First score and leftover.

There are three 1s and three 0s, so $F_0 = \ln \frac{3}{3} =$ **a)** -1 **b)** 0 **c)** 1. Then $p_0 = \sigma(0) =$ **a)** 0 **b)** 0.5 **c)** 1.

The leftover $r = y - p_0$ for points 1 to 6 is:

a) $-0.5, -0.5, 0.5, 0.5, -0.5, 0.5$ **b)** 0.5, 0.5, $-0.5, -0.5$, 0.5, -0.5 **c)** 0, 0, 1, 1, 0, 1

Step 2 Choose the split.

A good split puts the 1s on one side and the 0s on the other, as far as possible. For each split write the y values of the points that go left and right.

split	left leaf (y values)	right leaf (y values)
$x \le 2.5$	0, 0	1, 1, 0, 1
$x \le 3.5$	0, 0, 1	1, 0, 1
$x \le 4.5$	0, 0, 1, 1	0, 1

Best split:
a) $x \le 2.5$ **b)** $x \le 3.5$ **c)** $x \le 4.5$

Why is it best? *Its left leaf is all 0s and its right leaf is mostly 1s; the other splits mix 0s and 1s in both leaves.*

Step 3 Leaf values.

Every point has $p_0(1 - p_0) = 0.25$. For the best split fill in:

leaf	$\sum r$	$\sum p(1-p)$	leaf value $= \sum r / \sum p(1-p)$
left leaf	-1	0.5	-2
right leaf	$+1$	1	$+1$

Step 4 Add a small part, to the score.

Use $F_1 = F_0 + \nu \times$ leaf value, then $p_1 = \sigma(F_1)$ (use the σ values in the formula box), then $r_1 = y - p_1$ (round to 3 decimals).

point i	1	2	3	4	5	6
F_1	-1	-1	0.5	0.5	0.5	0.5
p_1	0.269	0.269	0.622	0.622	0.622	0.622
$r_1 = y - p_1$	-0.269	-0.269	0.378	0.378	-0.622	0.378

Step 5 What did the first tree miss?

Which point has the biggest leftover now (ignoring the sign)? **a)** point 1 **b)** point 3 **c)** point 5. **Why?** *Its true label is 0 but it sits next to points with label 1, so one stump gave it p = 0.622: the wrong direction.*

Think. Why do we add the leaf value to the score F, and not directly to the probability p? *A probability must stay between 0 and 1, but adding could push it above 1 or below 0. A score can be any number, and sigma squeezes it back into 0 to 1.*

(b) A point has $p = 0.5$ and $y = 1$. Is its leftover r **a)** positive **b)** negative, so does the next tree push its score **a)** up **b)** down?

Name: _____ **Roll number:** _____

AdaBoost, yes/no (classification)

Where you see a), b), c): circle the one right answer.

The idea. The answer is a class (*classification*). A *judge* is a stump: one question about x; it answers $+1$ or -1. We build a committee, one judge per round. After each round the points the judge got **wrong** count **more**, so the next judge pays attention to them. A judge with few mistakes gets a **louder voice** α. The committee's answer is the **sign** of the total vote.

point i	1	2	3	4	5	6
x_i	1	2	3	4	5	6
y_i	$+1$	$+1$	-1	-1	$+1$	-1

Three judges (each says $+1$ if $x \le t$, else -1)
A: $t = 2.5$ B: $t = 4.5$ C: $t = 5.5$

Formulas you need.
- mistake score ε = sum of the weights of the points the judge gets wrong
- voice $\alpha = \frac{1}{2} \ln \frac{1-\varepsilon}{\varepsilon}$
- new weight = old $\times e^{\alpha}$ (point was wrong) or old $\times e^{-\alpha}$ (point was right), then divide by the total so the weights add to 1
- vote $F(x) = \alpha_1 h_1(x) + \alpha_2 h_2(x)$, where $h_k(x) = \pm 1$ is what judge k answers; the committee's answer = $+1$ if $F > 0$, -1 if $F < 0$
- $\ln 5 = 1.609$, $\ln 4 = 1.386$, $e^{0.805} = 2.236$, $e^{-0.805} = 0.447$

Step 1 Pick the judge.

At the start every point has weight **a)** 1/2 **b)** 1/6 **c)** 1. So ε = (number of points wrong)/6. The best judge is the one with the smallest ε.

judge	rule	points it gets wrong	ε
A	$x \le 2.5 \to +1$, else -1	5	1/6
B	$x \le 4.5 \to +1$, else -1	3,4,5	3/6
C	$x \le 5.5 \to +1$, else -1	3,4	2/6

Best judge: **a)** A **b)** B **c)** C

$\varepsilon_1 =$ **a)** 1/6 **b)** 2/6 **c)** 3/6

Step 2 Give the judge a voice.

$\alpha_1 = \frac{1}{2} \ln \frac{1-\varepsilon_1}{\varepsilon_1} =$ **a)** 0.805 **b)** 1.609 **c)** 0.5

If $\varepsilon = \frac{1}{2}$ (a coin toss), then $\alpha =$ **a)** 0 **b)** 1 **c)** 0.5. **What does that tell you about such a judge's vote?** *A judge no better than guessing gets no vote, because it adds no information.*

Step 3 Change the weights.

Judge A got point **a)** 3 **b)** 4 **c)** 5 wrong. Work out the new weights yourself (3 decimals; tiny rounding differences are fine), using $e^{\alpha} = 2.236$ and $e^{-\alpha} = 0.447$ from the formula box.

	old weight $\times$ factor =	new weight (before dividing)
the wrong point	$\frac{1}{6} \times 2.236 =$	0.373
each right point (5 of them)	$\frac{1}{6} \times 0.447 =$	0.0745
total of all six	(wrong point) + 5$\times$ (right point) =	0.745
after dividing by the total	wrong point: 0.5	right point: 0.1

Why do we deliberately make the wrong point heavier? *So the next judge is chosen mainly to get that point right; the committee must cover what earlier judges missed.*

Step 4 Round 2.

Now point 5 has weight 0.5 and every other point 0.1. Find ε for each judge; add the weights of the points it gets wrong (same wrong points as Step 1).

judge	rule	wrong on points	ε
A	$x \le 2.5$	5	0.5
B	$x \le 4.5$	3,4,5	0.7
C	$x \le 5.5$	3,4	0.2

Best judge: **a)** A **b)** B **c)** C

$\alpha_2 =$ **a)** 0.693 **b)** 0.805 **c)** 1.386

Judge A was best in round 1. Why is it not chosen again? *A gets the heavy point 5 wrong, which now costs 0.5; the weights changed, so the best judge changed.*

Step 5 Take the vote.

$F(x) = 0.805\, h_1(x) + 0.693\, h_2(x)$ (h_1: judge A, h_2: judge C). For point 5:

$h_1(5) =$ **a)** $+1$ **b)** -1 $h_2(5) =$ **a)** $+1$ **b)** -1 $F =$ **a)** -0.112 **b)** $+0.112$ **c)** $+1.498$

The committee answers **a)** $+1$ **b)** -1. The true $y_5 = +1$, so the committee is **a)** right **b)** wrong.

Think. Why must every judge be better than a coin toss ($\varepsilon < \frac{1}{2}$)? *Otherwise alpha is zero or negative: the judge cannot help the committee.*

Name: _____ **Roll number:** _____

AdaBoost, numbers (regression)

Where you see a), b), c): circle the one right answer.

The idea. Same committee idea as page 3, but now a point is not just right or wrong: a tree can be wrong by a *little* or by a *lot*. So we measure **how** wrong each point is, compared with the worst mistake of the round. Points predicted badly get **relatively more weight** next round. A tree with a small average loss gets a **louder voice** α. The final answer is the **weighted median** of the trees' answers.

point i	1	2	3	4	5	6
x_i	1	2	3	4	5	6
y_i	2	2	2	6	6	10

Round 1: every weight is $\frac{1}{6}$. The best question is $x \le 3.5$: points 1 to 3 go to the left leaf, points 4 to 6 to the right leaf.

Formulas you need.
- leaf value h = average of y in the leaf (all weights are equal in round 1)
- error $\delta = |y - h|$; Δ = the largest δ of the round; relative loss $\ell = \delta/\Delta$ (between 0 and 1)
- average loss $\bar{\ell} = \sum w\,\ell$ (the weights add to 1); $\beta = \bar{\ell}\big/(1-\bar{\ell})$; voice $\alpha = \ln(1/\beta)$
- new weight $\propto w \times \beta^{\,1-\ell}$, then divide by the total so the weights add to 1
- weighted median: sort the answers, add up their α from the smallest answer, stop at the first answer where the running total reaches half of all the α
- $\ln 2 = 0.693$, $\sqrt{0.5} = 0.707$

Step 1 Grow the tree and measure the loss.

Work to 2 decimals; tiny rounding differences are fine.

Left leaf $h =$ 2 Right leaf $h =$ 22/3 ≈ 7.33 Largest error $\Delta =$ 8/3 ≈ 2.67

point i	1	2	3	4	5	6
y	2	2	2	6	6	10
h	2	2	2	7.33	7.33	7.33
$\delta = \lvert y - h \rvert$	0	0	0	1.33	1.33	2.67
$\ell = \delta/\Delta$	0	0	0	0.5	0.5	1

Step 2 Give the tree a voice.

$\bar{\ell} = \frac{1}{6}(\ell_1 + \cdots + \ell_6) =$ 1/3 $\beta = \bar{\ell}/(1-\bar{\ell}) =$ 1/2 $\alpha = \ln(1/\beta) =$ 0.693

A tree whose average loss $\bar{\ell}$ is close to $\frac{1}{2}$ (almost a coin toss) gets a **a)** loud **b)** quiet voice. **Why?** *Its predictions are not much better than guessing, so it should not influence the final answer.*

Step 3 Change the weights.

Multiplier = $\beta^{1-\ell}$ (a perfect point, $\ell = 0$, gets β^1; the worst point, $\ell = 1$, gets $\beta^0 = 1$). New weight = multiplier ÷ (sum of the six multipliers).

point i	1	2	3	4	5	6
ℓ	0	0	0	0.5	0.5	1
multiplier	0.5	0.5	0.5	0.707	0.707	1
new weight	0.128	0.128	0.128	0.181	0.181	0.256

sum of multipliers = 3.914

The point with the largest weight is **a)** point 1 **b)** point 4 **c)** point 6. **Why do the perfectly predicted points lose weight?** *The weights add to 1, so when the badly predicted points count more, the perfectly predicted ones must count less; the next tree is then pushed to fix point 6.*

Step 4 The final answer.

At $x = 6$ three trees answer as below. Find the weighted median.

tree	1	2	3
answer at $x = 6$	7.33	9.0	5.0
voice α	0.69	0.90	0.41

Sorted answers: 5.0, 7.33, 9.0

Running total of α: 0.41, 1.10, 2.0

Half of all $\alpha =$ 1.0

The weighted median is **a)** 5.0 **b)** 7.33 **c)** 9.0. **Why a median and not an average?** *A median ignores how extreme an answer is, so one tree with a wild answer cannot drag the final prediction away.*

Think. On page 3 a judge needs $\varepsilon < \frac{1}{2}$. What is the matching condition here? **a)** $\bar{\ell} < \frac{1}{2}$ **b)** $\bar{\ell} > \frac{1}{2}$. *Same reason: otherwise beta is 1 or more, alpha is zero or negative, and the tree cannot help.*